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Facile green synthesis of iron oxide nanoparticles via solid-state thermolysis of a chiral, 3D anhydrous potassium tris(oxalato)ferrate(III) precursor

A. Saritha^a, B. Raju^b, D. Narayana Rao^b, A. Roychowdhury^c, D. Das^c, K.A. Hussain^{a,*}^a Department of Physics, Kakatiya University, Warangal, Andhra Pradesh 506009, India^b School of Physics, University of Hyderabad, Hyderabad, Andhra Pradesh 500046, India^c UGC-DAE Consortium for Scientific Research, Kolkata 700098, India

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ABSTRACT

The iron oxide nanoparticles hematite (α -Fe₂O₃), maghemite (γ -Fe₂O₃), magnetite (Fe₃O₄) and wüstite (FeO) were synthesized via solid-state thermolysis of a chiral, 3D anhydrous potassium tris(oxalato)ferrate(III) (K₃[Fe(C₂O₄)₃]) precursor. A controlled heat treatment for the precursor material in air forms these four iron oxide polymorphs. Powder X-ray diffraction (XRD) and Raman spectroscopy were used to identify the different iron oxide nanoparticle polymorphs. The morphology of the obtained iron oxide nanoparticles was determined using a field emission scanning electron microscope (FE-SEM). The magnetic properties of the as-synthesized iron oxide nanoparticles was studied using a superconducting quantum interference device (SQUID), and the results indicated that they are weakly ferromagnetic, ferrimagnetic and paramagnetic in nature.

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1. Introduction

Iron oxide nanoparticles with different polymorph structures have been extensively studied due to their wide application in modern science and technology [1]. Currently, 16 iron oxide and oxyhydroxide polymorphs are known [2,3] and each polymorph has unique physical and chemical properties that are useful to wide variety of applications [4]. The most common iron oxide polymorphs are α -Fe₂O₃, γ -Fe₂O₃, Fe₃O₄ and FeO. The α -Fe₂O₃ phase has applications in catalysts, high-density magnetic storage media, pigments, anticorrosive agents, water splitting, water purification, solar energy conversion and gas sensors [5–8]. Because of their biocompatibility and non-toxicity, γ -Fe₂O₃ and Fe₃O₄ nanoparticles are used in magnetic resonance imaging (MRI) contrast agents, ferrofluids, targeted drug delivery, anticancer therapy, hyperthermia, cell labelling and separation, etc. [9–12]. The nonstoichiometric rock-salt structure of Fe_xO ($x = 0.84$ – 0.95) nanoparticles has great potential for catalytic and gas sensor applications [13,14]. The physical and chemical properties of iron oxide nanoparticles are greatly affected by the synthesis route and show significant diversity in terms of their oxidation state [15,16] and polymorphism.

Predicting the specific conditions to form a particular polymorph, stability and structural transformation is difficult [17]. Preparing a single phase polymorph for a specific application remains a challenge. Due to the wide variety of applications for the various iron oxide nanoparticle polymorphs, different synthetic procedures have been developed including the co-precipitation method, sonochemical method, sol-gel synthesis, thermal decomposition of organometallic compounds, etc. Of these methods, thermal decomposition of iron-containing precursors such as iron pentacarbonyl and iron-oleate is most commonly used to prepare iron oxide nanoparticles [18]. However, organometallic iron pentacarbonyl is toxic, volatile, expensive and limited to low-scale synthesis [19], whereas nanoparticles prepared using iron oleate precursor have low magnetization values [20]. In general, wet syntheses result in complex aqueous chemistries, such as hydrolysis, oxidation, and polymorphic transformations [21]. The solid-state thermolysis of iron-containing materials represents important precursors for preparing different iron oxides nanoparticle polymorphs. Therefore, transition metal ferrioxalates are potential precursors for synthesizing iron oxide nanoparticles and provides simple, non-toxic, cost effective and potentially large-scale production [22,23]. The synthesis and characterization of iron oxide nanoparticles via the solid-state thermolysis of iron-containing salts has been studied by many researchers [24–26]. However,

* Corresponding author. Tel.: +91 9848369406.

E-mail address: althaf.ku2@gmail.com (K.A. Hussain).

the formation mechanism of these iron oxide nanoparticles during thermolysis often differs and is influenced by many factors, such as the chemical composition of the precursor material, heating rate, and reaction atmosphere. The thermal decomposition of solid potassium tris(oxalato)ferrate(III) trihydrate has been extensively studied in several works [27–31]. However, the exact nature of the final compound is controversial, and moreover, no reports of the magnetic properties of the final residue were found. Recently, we reported an anhydrous potassium tris(oxalato)ferrate(III) structure [32,33] that is distinctly different from its hydrated counterpart [34], and a similar cubic enantiomorphous compound, called sodium potassium tris(oxalato)ferrate(III), was found in the literature with 1/6 of K⁺ ions replaced by Na⁺ ions [35,36]. These are the only compounds without crystalline water in the series of tris oxalato metalate(III) compounds.

Herein, we report a facile, nontoxic and inexpensive synthesis of four different iron oxide nanoparticle polymorphs via the solid-state thermolysis of an inorganic chiral, 3D anhydrous K₃[Fe(C₂O₄)₃] precursor material.

2. Experimental

2.1. Synthesis of iron oxide nanoparticles

The chiral, 3D anhydrous K₃[Fe(C₂O₄)₃] precursor complexes were synthesized according to the published procedure [32] and good quality crystals were grown via the slow evaporation method. To prepare the iron oxide nanoparticles, K₃[Fe(C₂O₄)₃] crystals were powdered and subsequently heated at various temperatures at a heating rate of 2°/min for 30 min in an electric furnace under ambient air before cooling to room temperature. The decomposition temperatures were selected using the thermogravimetric-differential thermal analysis (TG-DTA) data.

2.2. Characterization techniques

The TG-DTA measurements were performed using a Mettler Toledo instrument in the temperature range from room temperature to 1000 °C at a heating rate of 10 °C/min. The powder X-ray diffraction (XRD) patterns were recorded using JEOL-JDX-8P X-ray diffractometer fitted with a scintillation counter and nickel filtered Cu K α radiation (λ = 1.54056 Å). The iron oxide nanoparticle surface morphologies were determined using a field emission scanning electron microscope (FE-SEM) (Carl-Zeiss Ultra 55 model). The grain size distribution was measured by the dynamic light-scattering (DLS) method in water suspension using a particle size analyzer (Malvern Mastersizer 2000). Raman spectra were recorded at room temperature using a micro-Raman spectrometer (LABRAM-HR) with 514.5 nm laser excitation. Iron oxide nanoparticles rapidly oxidize during the laser power excitation; to avoid oxidizing the iron oxides, a laser power of < 1 mW was used throughout these experiments. The magnetic studies were performed using a superconducting quantum interference device (SQUID) magnetometer (Quantum Design, MPMS XL 7).

3. Results and discussion

The TGA-DTA curves obtained for the precursor material K₃[Fe(C₂O₄)₃] are shown in Fig. 1.

The TGA curve shows three separate decomposition stages. An initial mass loss of 24% (cal. 23.3%) was observed between 294 and 330 °C, and the corresponding DTA curve shows a sharp exothermic peak at 324 °C.

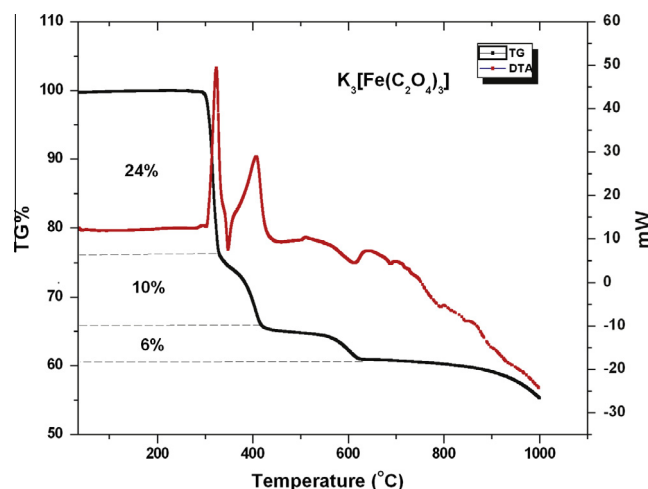
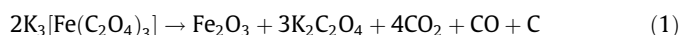
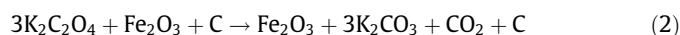
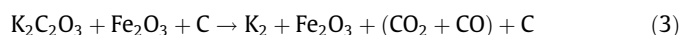


Fig. 1. TG-DTA curves for the K₃[Fe(C₂O₄)₃] precursor.

A mass loss of 10% (cal. 10.06%) was observed in the second stage between 330 and 414 °C, and the corresponding exothermic DTA peak is centered at 407 °C.



The third decomposition stage, which had a mass loss of 6% (cal. 6.4%), was observed between 414 and 618 °C, and the corresponding small exothermic peak in the DTA occurred at 528 °C.



The main aim of the present work is to prepare iron oxide nanoparticles via the solid-state thermolysis of a K₃[Fe(C₂O₄)₃] precursor material. The calcined samples were expected to contain potassium carbonate impurities. To remove these impurities, the decomposed species (except FeO particles) were washed with deionized water and methanol and separated using a magnet. The species were then dried in an oven for 5 h at 50 °C and characterized. The XRD patterns of the anhydrous K₃[Fe(C₂O₄)₃] and its decomposition products at various temperatures are shown in Fig. 2(a) through (f).

The room temperature XRD patterns for the anhydrous K₃[Fe(C₂O₄)₃] precursor compound (Fig. 2a) indicate a cubic system with an enantiomorphous P4₁32 space group [32,33]. The XRD patterns of the decomposition products at various temperatures (Fig. 2b–f) clearly observed that indicate all of the diffraction peaks for the precursor material disappeared, while new peaks appeared. The diffraction peaks in the samples annealed at 370 and 550 °C (Fig. 2b and d) appear with 2 θ values of 33.05, 35.66, 40.20, 49.47, 54.84, 62.80 and 69.47. These values closely match the α -Fe₂O₃ phase (JCPDS No. 89-0597). The samples calcined at 450 °C (Fig. 2c) appeared red-brown in color and exhibit diffraction peaks at 33.35, 34.20, 35.00, 38.73, 41.93, 43.94, 44.38, 46.39, 49.47, 58.96 and 66.04, which closely match the values for the γ -Fe₂O₃ phase (JCPDS No. 39-1436). FeO forms at 650 °C (Fig. 2e) with diffraction peaks at 35.41, 36.28, 41.02 and 59.01, which is consistent with the literature values (JCPDS No. 89-0690). By selectively choosing the temperatures for the precursor decomposition, three different polymorphs of iron oxide nanoparticles, α -Fe₂O₃, γ -Fe₂O₃, and FeO, were obtained. In contrast, keeping the precursor material powder in a furnace already heated to 400 °C caused the powder to become black within 5 min and yielded a diffractogram (Fig. 2f) with 2 θ values of 33.05, 35.66, 40.20, 49.47, 54.84, 62.80 and 69.47. These values are consistent with the Fe₃O₄ phase (JCPDS No. 19-0629). Although the decomposed species were washed to

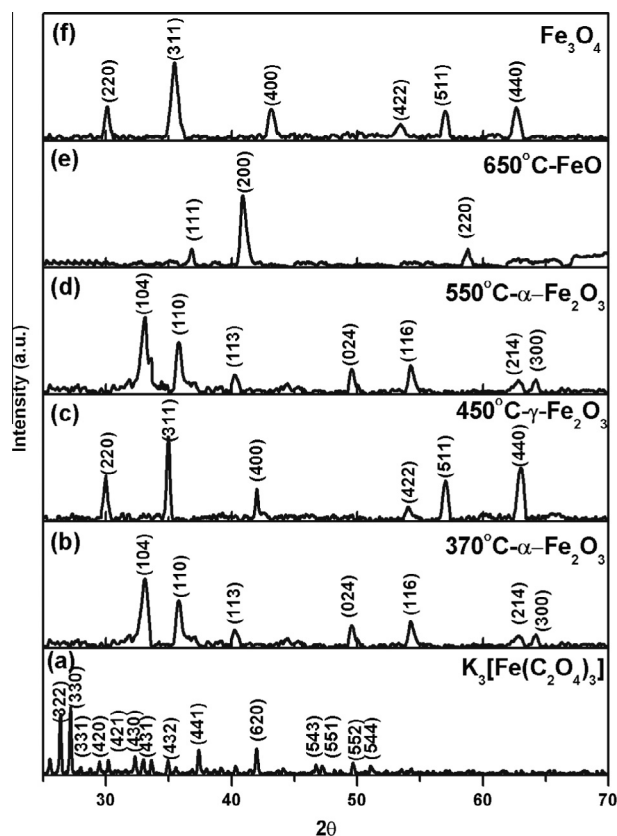


Fig. 2. XRD patterns for anhydrous $K_3[Fe(C_2O_4)_3]$ and its decomposition products for various temperatures.

remove impurities, additional low-intensity peaks appeared in the diffractograms because of residual precursor material impurities.

The controlled heat treatment of the $K_3[Fe(C_2O_4)_3]$ precursor in air formed in four different iron oxide nanoparticle polymorphs, α - Fe_2O_3 , γ - Fe_2O_3 , Fe_3O_4 and FeO , which are shown in Fig. 3.

However, the γ - Fe_2O_3 and Fe_3O_4 phases could not be distinguished from the XRD patterns because their peaks appear almost at the same positions. Moreover, we cannot estimate the quantitative and qualitative phase composition of the calcination products using the X-ray diffraction results alone. Nevertheless, spectroscopic approaches now include more sophisticated

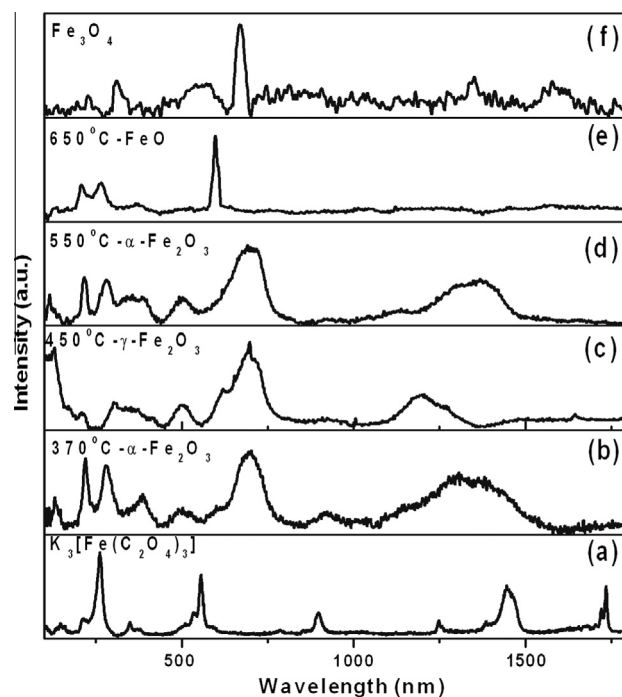


Fig. 4. Raman spectra of (a) $K_3[Fe(C_2O_4)_3]$, (b, d) α - Fe_2O_3 phase, (c) γ - Fe_2O_3 phase, (e) FeO phase and (f) Fe_3O_4 phase.

characterization tools for identifying material phase compositions. Raman spectroscopy is an important tool for distinguishing different iron oxide nanoparticle polymorphs because of the so-called phonon confinement effects; different iron oxide nanoparticle polymorphs exhibit distinct Raman signatures [34].

The Raman spectra of the $K_3[Fe(C_2O_4)_3]$ precursor and the decomposition products are shown in Fig. 4(a)–(f).

Fig. 4(a) shows the room temperature Raman spectra for the $K_3[Fe(C_2O_4)_3]$ precursor, which exhibits characterization peaks at 136, 260, 350, and 555 cm^{-1} with the $\nu(Fe-O)$ and $\delta(C-O)$ modes at 780 cm^{-1} and the $\nu(C-O)$ mode exhibited at 1246, 1445, and 1732 cm^{-1} [37,38]. The Raman signatures for the material decomposed at 370 and 550 $^{\circ}C$ (Fig. 3b and e) exhibit peaks at 227 (A_{1g}), 293 (E_g), 405 (E_g), 494 (A_{1g}), which agree well with the α - Fe_2O_3 phase [39], and the Raman peak other than these symmetry

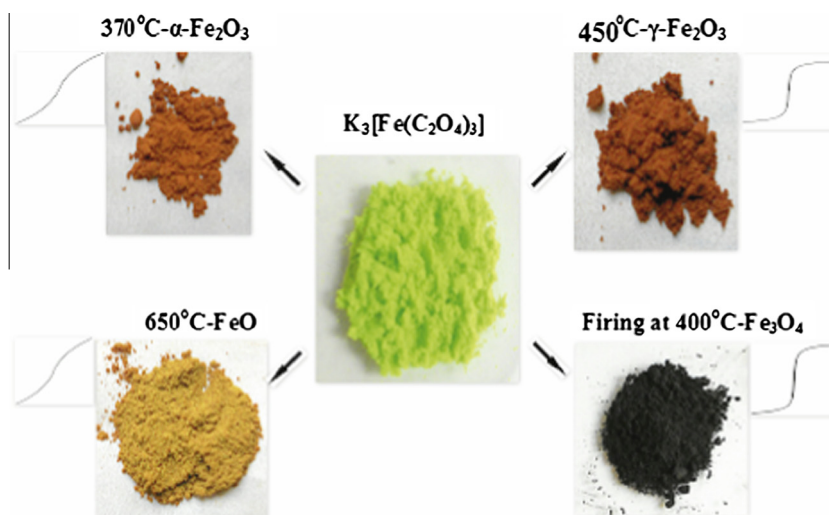


Fig. 3. α - Fe_2O_3 , γ - Fe_2O_3 , Fe_3O_4 and FeO obtained from the $K_3[Fe(C_2O_4)_3]$ precursor using different decomposition temperatures.

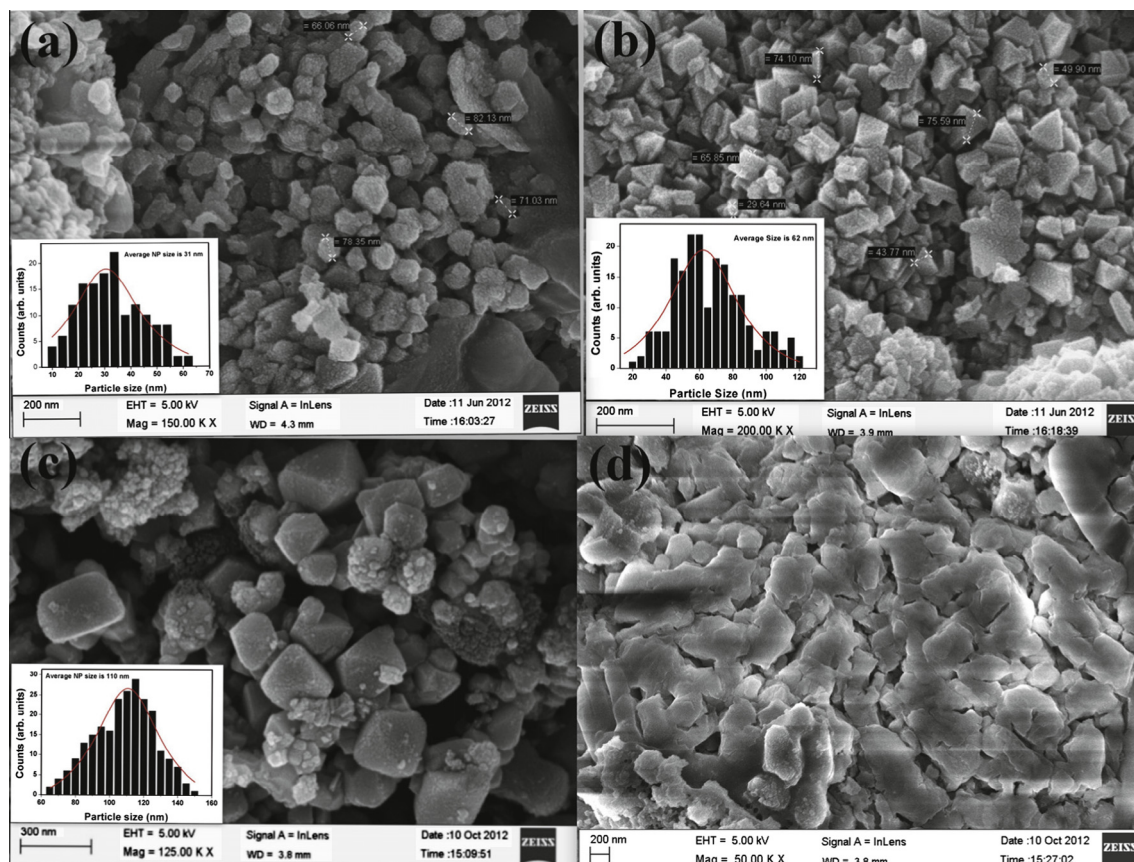


Fig. 5. FE-SEM images of (a) α - Fe_2O_3 , (b) γ - Fe_2O_3 , (c) Fe_3O_4 , (d) FeO nanoparticles obtained from the $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ precursor material at various decomposition temperatures.

allowed modes at $\sim 700\text{ cm}^{-1}$ is attributed to nanosized α - Fe_2O_3 particles [40]. The material decomposed at 450°C (Fig. 4c) exhibits three Raman signatures at 350 cm^{-1} (E_g), 494 cm^{-1} (T_{2g}) and 692 cm^{-1} (A_{1g}) that are characteristic of the inverse spinel γ - Fe_2O_3 phase [41]. Fig. 4(e) shows the Raman signatures for the material obtained at 650°C , which exhibits peaks at 215 , 278 , 388 and 594 cm^{-1} indicative of the FeO phase [42]. The spinel magnetite phase was confirmed by the Raman signatures at (Fig. 4f) 310 cm^{-1} (T_{2g}), 554 cm^{-1} (T_{2g}) and 672 cm^{-1} (A_{1g}) [41].

The size and morphology of the four different iron oxide nanoparticle polymorphs prepared via this solid-state thermolysis process were observed using FE-SEM and shown in Fig 5(a)–(d).

The FE-SEM images indicate the different iron oxide nanoparticle polymorphs obtained at various decomposition temperatures possess differing morphologies. The α - Fe_2O_3 particles (Fig. 5a) obtained at 370°C exhibit a spherical morphology with a corresponding average grain size of approximately 20 – 40 nm . The γ - Fe_2O_3 particle morphology (Fig. 5b) are octahedral, and the average grain size was 30 – 70 nm . The Fe_3O_4 particles (Fig. 5c) are also octahedral; however, their particle size increased to 100 – 150 nm . Irregular shapes and sizes (Fig. 5d) were observed for the FeO particles due to their agglomeration at higher temperatures.

The room temperature field dependencies of the magnetization curves (hysteresis loops) for the four different iron oxide nanoparticle polymorphs and the precursor material under an applied field of $\pm 10\text{ kOe}$ are shown in Fig. 6.

The $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ precursor is paramagnetic at room temperature [32,33]. The hysteresis loops for the α - Fe_2O_3 nanoparticles (Fig. 6) obtained at 370 and 550°C did not reach saturation and show a weak parasitic ferromagnetism because their electron spin

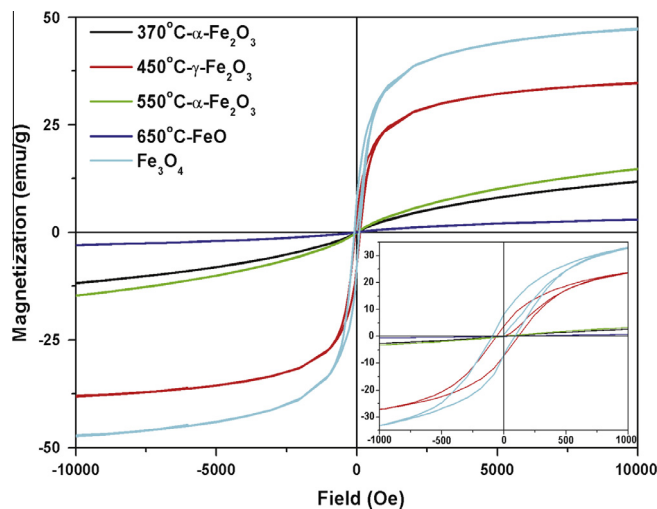


Fig. 6. Room temperature M – H curves for α - Fe_2O_3 , γ - Fe_2O_3 , FeO and Fe_3O_4 particles obtained at various decomposition temperatures.

magnetic moments were not in a rigid anti-parallel arrangement [43]. The corresponding remanent (M_r) and coercive forces (H_c) are 0.086 emu/g and 12.7 Oe , respectively. The γ - Fe_2O_3 nanoparticles obtained at 450°C show a ferrimagnetic behavior with a saturation magnetization (M_s) of 34 emu/g and corresponding M_r and H_c of 2.87 emu/g and 92 Oe , respectively. The M_s and H_c of the γ - Fe_2O_3 nanoparticles were much lower than those of the bulk γ - Fe_2O_3 (76 emu/g and 300 Oe), which may be attributed to their nano scale dimensions and surface effects [44]. The octahedral

Fe₃O₄ nanoparticles exhibit ferrimagnetic behavior with $M_s = 47$ emu/g, $H_c = 94$ Oe and $M_r = 8.68$ emu/g. These values are smaller than for bulk Fe₃O₄. However, similar magnetization values were observed for Fe₃O₄ nanoparticles obtained via a solvent free synthesis [45]. The FeO nanoparticles (Fig. 6) obtained at 650 °C exhibits a paramagnetic behavior.

The above results indicate calcining the precursor powder at 370 °C, forms thermodynamically stable α -Fe₂O₃ nanoparticles with a spherical morphology. In contrast, setting the calcination temperature to 450 °C yields octahedral spinel metastable γ -Fe₂O₃ nanoparticles 40–70 nm in size. The observed phase transition from α -Fe₂O₃ to γ -Fe₂O₃ is due to the changed particle size and morphological modification from spherical to octahedral. The critical particle size for the phase transition from α -Fe₂O₃ to γ -Fe₂O₃ via a dry synthesis is above 40 nm, and these observations corroborate an earlier work on iron oxide polymorphs [40]. When the precursor material is calcined at 550 °C, α -Fe₂O₃ nanoparticles were again formed. Thermally induced polymorphic transitions are known to commonly occur during iron oxide nanoparticle heat treatment syntheses [46]. The observed polymorph transformation from the thermally unstable γ -Fe₂O₃ to the stable α -Fe₂O₃ at high temperatures (550 °C) is proposed due to the tetrahedral defects that occur during the heat treatment synthesis [47]. The FeO phase is detected when the calcination temperature is 650 °C. The Fe₃O₄ nanoparticles formed when the precursor materials are fired at 400 °C and repeated attempts yield the same result.

4. Conclusion

In summary, four different iron oxide nanoparticle polymorphs, α -Fe₂O₃, γ -Fe₂O₃, Fe₃O₄ and FeO, were prepared via solid-state thermolysis of a chiral, 3D anhydrous K₃[Fe(C₂O₄)₃] precursor. The proposed method is fascinating because it offers a simple, non-toxic, inexpensive and potentially large-scale production of four important iron oxide nanoparticle polymorphs. This process yields weak ferromagnetic spherical α -Fe₂O₃ particles at 370 °C, with an average particle size of 30 nm, and ferrimagnetic octahedral γ -Fe₂O₃ particles at 450 °C, with particle sizes ranging from 30 to 70 nm. Paramagnetic FeO particles with irregular shapes and sizes were observed at 650 °C. Ferrimagnetic Fe₃O₄ nanoparticles were obtained by placing the precursor in a furnace already heated to 400 °C, and the resultant octahedral particles were 100–150 nm in sizes.

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